

CAB203 Graphs Project (Assessment 3)

Rory Blair: n10773398

Linear Algebra → blendWheat(csvfilename)

1). Mathematical Explanation

The objective of this project is to decide how to dispense and blend the wheat, from three different bins, to optimise the total of high protein grade wheat that can be sold. There are three requirements that need to be met, so that the objective can be completed. The 1st requirement is that there needs to be 14% of protein by weight as a minimum in the final blend result. The 2nd requirement is that there needs to be 12.5% of moisture by weight as a maximum in the final blend result. The 3rd requirement is that the sum of the weights of wheat from all bins included in the blend equals the total weight of the blend. The input given is a '.csv' file called 'csvfilename'. This file includes the separate bins and their number of certain segments, such as weights, protein and moisture. These requirements can be seen as quantified predicates below:

$$\forall bin_i \in Bins: Weight(bin_i, Bins_{weight}) \wedge Blend(bin_i, blend_{amount}) \rightarrow Bins_{weight} = blend_{amount}$$

$$\sum_{bin_i \in Bins} Protein(bin_i, Protein_Scale) \times Blend(bin_i, blend_{amount}) \geq 14 \times blend_{amount}$$

$$\sum_{bin_i \in Bins} Moisture(bin_i, Moisture_Scale) \times Blend(bin_i, blend_{amount}) \leq 12.5 \times blend_{amount}$$

To solve this problem multiple functions from the python library called 'numpy' or 'NumPy' will be used. 'numpy' is a library that is usually used for scientific calculations, providing complex array objects, matrices, and array of procedures that can be used on arrays. These procedures include basic linear algebra, random simulation, discrete Fourier transforms, mathematical, logical, shape manipulation and sorting. The functions being used from the 'numpy' library are 'array', 'dot', 'inv', 'min' and 'abs'.

The 'array' function is being used to create an 'ndarray', which is able to be used with the other functions within the 'numpy' library. The input for this function is another python array, whilst the output is an 'ndarray'. The 'dot' function is being used to calculate the solution for how much each bin needs to contribute towards the final blend by using multiplication. The input for this function is two 'ndarray' variables and the output is also an 'ndarray', which is the dot product of the two inputs. The 'inv' function is used to easily inverse a matrix. The input is an 'ndarray' and the output is the inverse version of the input. The 'min' function is used to get the minimum value from an 'ndarray' given to it. The input is an 'ndarray' and the output is a single value, which is the minimum value within the input array. The 'abs' function is used to find the absolute value of a integer given to it. The input is a single integer, and the output is the same integer without a negative value, if the input was negative.

2). Mathematical Solution

The solution starts with being able to read the '.csv' file given, which is opened and a 'DictReader' from the csv library is used to read through and assert values to their respective bins. During this process the dictionary being made is then turned into a list, which is easier to convert to 'numpy' arrays needed for later during the solution. After reading through the csv file, the program then splits the list into separate 'ndarray' variables based on Weights, Protein and Moisture.

Then the program creates the required weights variable, which would all be the integer 1. The program also creates a variable 'H' which is the initial amount of protein, moisture and weight, but the weight is replaced with the required weight variable. The program then creates a variable called 'N', which is the needed values for the protein, moisture and weight in the form of an 'ndarray' [14 12.5 1]. Then the variable, 'X', is calculated multiplying the inverse of the variable 'H' and the variable 'N', which figures out how much from each bin is required.

The program then calculates the scaling factor for the bins by finding the minimum value within the result of Weights, 'W', divided by the solution, 'X'. The program then calculates the blending amount. The program then formats the output of the blending amount into a dictionary format. The program finally finds the total weight of the blending amount by summing up all the weights from each bin used. Three test cases are seen below in table 1.

	Test case 1	Test case 2	Test case 3
Csvfilename	Bins1.csv	Bins5.csv	Bins9.csv
Bins	[{'Bin': 'A', 'Weight': '12', 'Protein': '15', 'Moisture': '12.5'}, {'Bin': 'B', 'Weight': '15', 'Protein': '13.5', 'Moisture': '12'}, {'Bin': 'C', 'Weight': '7', 'Protein': '12', 'Moisture': '14'}]	[{'Bin': 'A', 'Weight': '22', 'Protein': '15', 'Moisture': '12.5'}, {'Bin': 'B', 'Weight': '15', 'Protein': '13.5', 'Moisture': '12.5'}, {'Bin': 'C', 'Weight': '4', 'Protein': '1', 'Moisture': '14'}]	[{'Bin': 'Top', 'Weight': '22', 'Protein': '15', 'Moisture': '12'}, {'Bin': 'Middle', 'Weight': '15', 'Protein': '13.5', 'Moisture': '12'}, {'Bin': 'Bottom', 'Weight': '4', 'Protein': '12', 'Moisture': '14'}]
Weights	[12. 15. 7.]	[22. 15. 4.]	[22. 15. 4.]
Protein	[15. 13.5 12.]	[15. 13.5 1.]	[15. 13.5 12.]
Moisture	[12.5 12. 14.]	[12.5 12.5 14.]	[12. 12. 14.]
Have	[[15. 13.5 12.] [12.5 12. 14.] [1. 1. 1.]]	[[15. 13.5 1.] [12.5 12.5 14.] [1. 1. 1.]]	[[15. 13.5 12.] [12. 12. 14.] [1. 1. 1.]]
Need	[14. 12.5 1.]		
Solution	[0.46666667 0.4 0.13333333]	[3.33333333e-01 6.66666667e-01 -1.77635684e-15]	[0.58333333 0.16666667 0.25]
Scale	25.714285714285708	22.499999999999984	16.000000000000057
Blend Amount	[12. 10.28571429 3.42857143]	[7.50000000e+00 1.50000000e+01 -3.99680289e-14]	[9.33333333 2.66666667 4.]
Final Return	({'A': 12.0, 'B': 10.29, 'C': 3.43}, 25.71)	({'A': 7.5, 'B': 15.0, 'C': 0.0}, 22.5)	({'Top': 9.33, 'Middle': 2.67, 'Bottom': 4.0}, 16.0)

Table 1: Test cases for blendWheat(csvfilename)

3). Python Implementation

There are many variables that are made throughout the program. These variables include 'BD', 'W', 'P', 'M', 'WW', 'H', 'N', 'X', 's', 'BA', 'BN', 'BL' and 'a'. The variable 'BD' is the acronym for 'Bin Dictionary', which is a list that contains the data from the dictionary created using the csv reader. The variable 'W' is the separate weights from all three bin included into an 'ndarray'. The variable 'P' is the separate protein from all three bins included into an 'ndarray'. The variable 'M' is the separate moisture from all three bins included into an 'ndarray'. The variable 'WW' is the required weight, which includes the integer 1 three separate times.

The variable 'H' is the initial protein, moisture and required weight in a 3x3 matrix, which is represented as nested 'ndarray' variables. The variable 'N' is an 'ndarray' that the required protein, moisture and weight that the final solution must equal. The variable 'X' is an 'ndarray' that is the solution where the inverse of variable 'H' is multiplied with the variable 'N' to find the solution to the problem. The inverse is used as the inverse of the initial data multiplied by the data needed calculates a solution. (Said-Houari, 2017) The variable 's' is an integer that represents the scaling factor that will be used to find the blend of all three bins, which is calculated by finding the minimum value from the division of variable 'X' by variable 'W'. Using these two variables, 's' and 'X', and multiplying them together, the variable 'BA' or blend amount can be calculated, which is an 'ndarray' (McKague, CAB203 Discrete Structures Lecture 11: Linear Algebra, 2024).

The variable 'BN' or bin names gets the names of each bin into a list, so that it is able to be formatted into the final result. The variable 'BL' is used to format the variables into a dictionary, where the 'key' values are the bins names, and the 'item' values are the required amount from each bin into the blend. The variable 'a' is an integer that represents the total weight of the final blend, which is calculated by adding all the 'item' values within the dictionary.

References

McKague, M. (2024, May 13). *CAB203 Discrete Structures Lecture 11: Linear Algebra*. Retrieved from QUT: https://canvas.qut.edu.au/courses/16665/pages/11-dot-2-lecture-material?module_item_id=1552001

Said-Houari, B. (2017). *Linear Algebra*. Birkhäuser Cham.